

JONATHAN GAYNOR

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EDUCATION

Columbia University M.S. in Mechanical Engineering Relevant Coursework: Finite Element Analysis (ANSYS), Spaceflight Mechanics, MEMS, Mechatronics.	New York, NY Expected Dec 2027
Grinnell College Bachelor of Arts Bachelor of Arts in General Science, Concentration in Mathematics.	Grinnell, IA May 2026
Penn State University No Degree Earned Coursework in Electrical-Mechanical Engineering Technology.	Lemont Furnace, PA May 2024

PROJECTS

Catheter Instrumentation Machine Lead Systems Developer - Received a \$2,500 grant to develop a low-cost medical validation apparatus with load-cell force measurement and thermal instrumentation arrays, reducing production costs from over \$1,000 to under \$25. - Programmed feedback loops in Arduino IDE to process multi-sensor data streams and execute precision testing intervals.	Jun 2024 - Jun 2024
Autonomous Line-Following Vehicle Hardware Co-Designer - Optimized optoelectronic sensor array layout, refining double LED spacing to improve error-correction tracking margins. - Created a circuit diagram and soldered 10+ components to consolidate sensor array with the vehicle's control system.	Apr 2024 - Apr 2024
Multi-Joint Mechatronic Robotic Arm Mechanical Design Lead - Modeled multi-DOF robotic kinematics in SolidWorks, validating link dimensions through geometric motion testing. - Fabricated 20+ 3D-printed PLA components using failure analysis to optimize infill densities and center-of-mass balancing. - Integrated 3 high-torque actuators with microprocessor control boards to synchronize mechanical, electrical, and control loops.	Nov 2023 - Dec 2023

WORK EXPERIENCE

Helping Hands Mechanical Engineering Volunteer - Designed and manufactured 3D-printed prosthetic arms for underprivileged children, assembling 20+ precision components per arm across 15+ devices delivered as custom-fit solutions. - Collaborated on 3D-printed face shield fabrication during COVID-19, supporting donation and delivery to Jackson Memorial Hospital in Miami, FL.	Miami Beach, FL Aug 2020 - Present
Gaynor & Company Inc IT Advisor and Customer Service Assistant Delivered IT advisory and technical troubleshooting, resolving 50+ hardware and software issues to maintain operational efficiency.	Hollywood, FL Jun 2022 - Present

SKILLS

Embedded Systems & Programming: C/C++, Python (NumPy, Pandas), MATLAB, Arduino IDE, Java, Jupyter Notebook.
Design & Simulation: SolidWorks, Fusion 360, AutoCAD, FreeCAD, Tinkercad.
Fabrication & Additive Manufacturing: SLA/FDM 3D Printing, PrusaSlicer, Ultimaker Cura, Multi-Material Printing, Mechanical Assembly.

ACTIVITIES & LEADERSHIP

NCAA Collegiate Baseball Left-Handed Pitcher and Team Leader Played and led teams to victory, making conference playoffs or regionals in all 4 seasons.	Grinnell College, Penn State University Aug 2022 - May 2026
Aeronautic Learning and Engineers Club Member Volunteered in student-led AutoCAD learning and object designing.	Grinnell, IA Jan 2026 - May 2026
Maker's and 3D Printing Clubs Member Specialized in advanced slicing toolpath and multi-material printing to develop 5+ custom prints for fun.	Lemont Furnace, PA Sep 2022 - May 2024